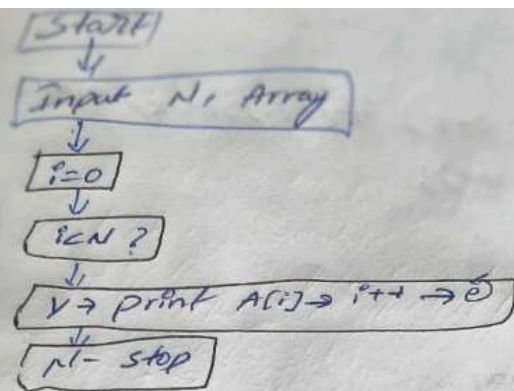


COS AT3

flowchart to code
Conversion

Name: K. Anil Kumar Reddy
Reg no: 192525239
Course Name: Data structure
Course code: CSA0306

Luxor Fine Writer 05

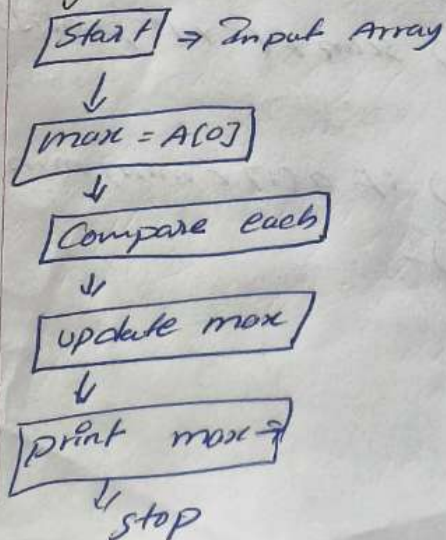


Code

```

int main() {
    int a[50], n, i;
    scanf("%d", &n);
    for (i=0; i<n; i++)
        scanf("%d", &a[i]);
    for (i=0; i<n; i++)
        printf("%d", a[i]);
}
  
```

2. Largest:

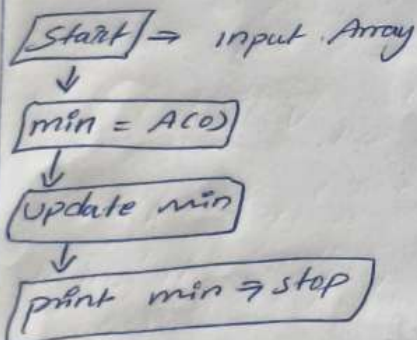



```

int main() {
    int a[50], n, i, max;
    scanf("%d", &n);
    for (i=0; i<n; i++) scanf("%d", &a[i]);
    max = a[0];
    for (i=1; i<n; i++) if (a[i] > max) max = a[i];
    printf("%d", max);
}

```

3. Smallest:-

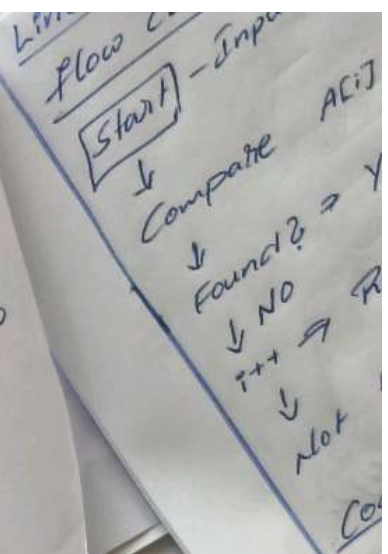


Code:-

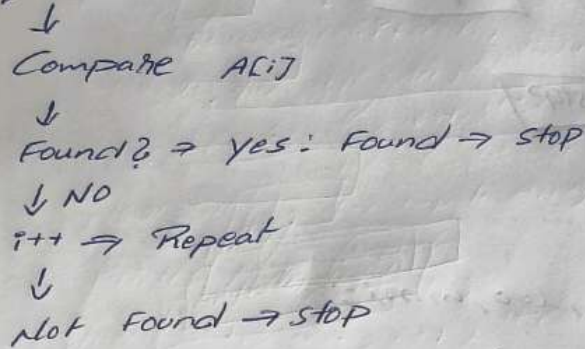
```

int main() {
    int a[50], n, i, min;
    scanf("%d", &n);
    for (i=0; i<n; i++) scanf("%d", &a[i]);
    min = a[0];
    for (i=1; i<n; i++) if (a[i] < min) min = a[i];
    printf("%d", min);
}

```



Start - Input Array, key

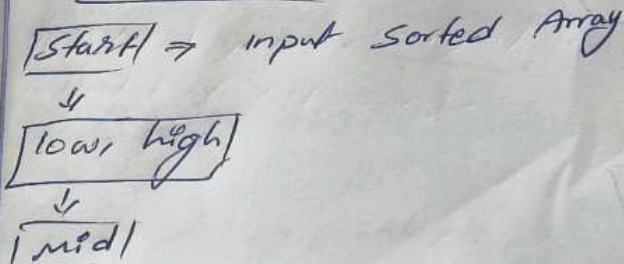
Code

```

int main() {
    int a[50], n, i, k;
    scanf("%d", &n);
    for (i=0; i<n; i++) scanf("%d", &a[i]);
    scanf("%d", &k);
    for (i=0; i<n; i++) if (a[i] == k)
    { printf("Found"); return 0; }
    printf("Not Found");
}

```

Flow Chart



Equal? $\rightarrow$ Found

$\downarrow$ No

Update low/high $\rightarrow$ Repeat

$\downarrow$

Not Found $\rightarrow$ Stop

Code:-

```
int main() {
```

```
    int a[50], n, k, l=0, h, m;
```

```
    scanf("%d", &n)
```

```
    for (i=0; i<n; i++) scanf("%d", &a[i]);
```

```
    scanf("%d", &k); h=n-1;
```

```
    while (l<=h) {
```

```
        m = (l+h)/2
```

```
        if (a[m] < k) l=m+1; else h=m-1;
```

```
    }
```

```
    printf("not Found");
```

```
}
```

6. Insert at position

Start $\rightarrow$ Input

$\downarrow$

Shift elements

$\downarrow$

Insert x

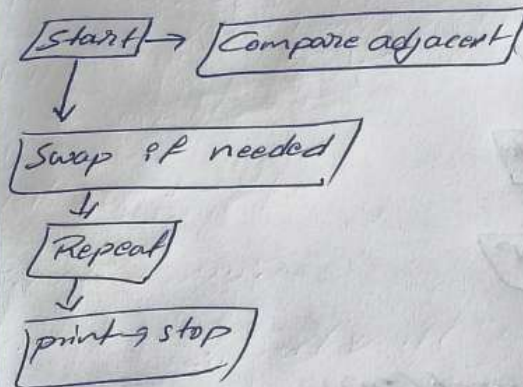
$\downarrow$

Print $\rightarrow$ Stop

Code

```
a = [12, 13, 14]
for (i=n; i>p; i--)
    a[i] = a[i-1];
    a[p-1] = x;
```

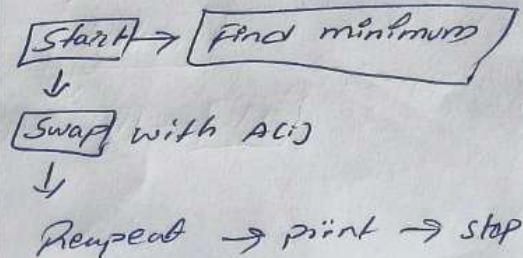
7. Bubble sort



Code

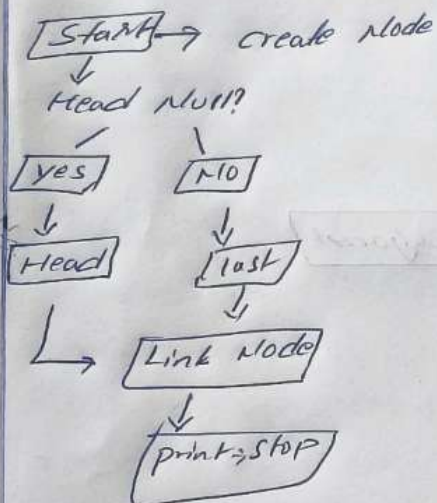
```
for (i=0; i<n-1; i++)
    for (j=0; j<n-i-1; j++)
        if (a[j] > a[j+1]) {
            t = a[j]; a[j] = a[j+1]; a[j+1] = t;
        }
    }
```

8. Selection sort



9. Singly Linked List

Flow chart



Code:-

```

n = malloc(sizeof(*n));
n->data = x; n->next = NULL;
if (!h) h = n
else {
  t = h;
  while (t->next) t = t->next;
  t->next = n;
}
  
```

10. Kruskal Algorithm

```

Start
+
Input edges
↓
sort weight
↓
Cycle
  ✓   ↓   ✗
yes  no
skip add
  ✓   ✗
Repeat & stop.
  
```

Code:

```
#include <stdio.h>
struct t { int u, v, w; };
```

```
int main() {
```

```
    struct E e[20], t; int
    n, m, i, j, p[20], c=0
```

```
    for (c=0; c<m-1; c++)
```

```
        for (i=0; i<m; i++) {
```

```
            a=e[i].u, b=e[i].v,
```

```
            if (p[a] != p[b]) {
```

```
                printf("%d", a, b)
```

```
                p[a]=b;
```

```
            }
```

```
        }
```